Tinea Infections

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Epidemiology and Etiology

- ☑ Tinea infections are known as dermatophytosis.
- ☑ Dermatophytosis is found worldwide and accounts for 8–10% of skin outpatient department (OPD).
- ☑ The most common in children is tinea capitis, commonly seen in urban preschool-aged and school-aged children.
- ☑ In Southern India, 60–80% cases of tinea capitis are caused by *Trichophyton violaceum*.
- ☑ Tinea pedis, manuum, corporis, barbae, and cruris occur mainly in adolescents.
- ☑ *Mode of spread*: Contact with infected person or animal and infected fomites (combs, brushes, caps, hats, couches, floors of swimming baths, etc.).

Tinea Infection: Dermatophytes

Causative agents	*Trichophyton*—causes infections on skin, hair, and nails	*Microsporum*—causes infections on skin and hair	*Epidermophyton*—causes infections on skin and nails
Types	☑ Tinea corporis (skin) ☑ Tinea manuum/pedis (hands/feet) ☑ Tinea capitis (scalp) ☑ Tinea barbae (bearded areas) ☑ Tinea unguium (nail)	Most common is tinea capitis	

Tinea Capitis

Tinea infection is common between ages 2 and 9 years living in crowded areas.

Symptoms

- ☑ Diffuse itchy scaling of the scalp resembling dandruff
- ☑ Patches of hair loss
- ☑ Spreads through combs, pillows, hats, etc.

Treatment

- ☑ Topical application of shampoo with 2.5% selenium sulfide/zinc pyrithione/ketoconazole
- ☑ Oral griseofulvin (microcrystalline—20–25 mg/kg/day maximum 1,000 mg or ultramicrosize—10–15 mg/kg/day maximum 750 mg) for 6 weeks
- ☑ Oral terbinafine at 3–6 mg/kg/day for 4–6 weeks can be used in cases of noninflammatory tinea capitis in 4 years and older

Tinea Corporis

It involves the nonhairy parts of the body such as face and arms. Single to multiple large, itchy erythematous, well-defined circinate or circular lesions with border studded with papules, vesicles, pustules or crusts with a central clearing.

Treatment

- ☑ *Mild-to-moderate:* Topical antifungal agents (imidazole, terbinafine, butenafine, and naftifine) twice daily for 2–4 weeks.
- ☑ Oral itraconazole for 1–2 weeks can be added if more than two sites involved.
- ☑ *Extensive involvement cases:* Oral griseofulvin for 2–8 weeks or oral terbinafine for 2–6 weeks along with topical antifungal agents

Tinea Pedis

It usually occurs in adolescents and is rare prior to puberty. It is caused by *Trichophyton rubrum*, *Trichophyton interdigitale*, and *Epidermophyton floccosum*.

Types

- ☑ *Interdigital tinea pedis:* Pruritic erosions or scales between the toes, especially in the third and fourth digital interspaces. Interdigital fissures may cause pain.
- ☑ *Hyperkeratotic (moccasin) tinea pedis:* Diffuse, erythematous, hyperkeratotic eruption involving the soles and medial and lateral surfaces of the feet, resembling a "moccasin" distribution.
- ☑ *Vesiculobullous tinea pedis:* Pruritic, sometimes painful, vesicular, or bullous eruption.

Tinea Pedis

Treatment

- ☑ Avoidance of occlusive footwear, drying between the toes after bathing and use of absorbent antifungal powder (zinc undecylenate).
- ☑ Topical therapy with imidazole, allylamines, and tolnaftate for 2–4 weeks.
- ☑ Oral griseofulvin (microsized) at 10–20 mg/kg/day for 6–8 weeks.
- ☑ Oral terbinafine at 62.5 mg/day (<20 kg), 125 mg/day (20–40 kg), and 250 mg (>40 kg) for 2 weeks.
- ☑ Hyperkeratotic tinea pedis may benefit from combining antifungal treatment with a topical keratolytic, such as salicylic acid.

Tinea Manuum

- ☑ Tinea manuum is a dermatophyte infection of the hand.
- ☑ Patients present with a hyperkeratotic eruption on the palm or annular plaques similar to tinea corporis on the dorsal hand.
- ☑ Tinea manuum commonly occurs in association with tinea pedis and is often unilateral.
- ☑ This clinical presentation is often referred to as "two feet-one hand syndrome".
- ☑ Treatment is similar to treatment of tinea pedis.

Tinea Cruris

Tinea cruris (also known as jock itch) is a dermatophyte infection involving the crural fold.

Etiology

T. rubrum, E. floccosum, and *T. interdigitale.*

Predisposing Factors

Copious sweating, obesity, diabetes, and immunodeficiency.

Tinea Cruris

Clinical Features

- ☑ Erythematous or hyperpigmented patch on the proximal medial thigh. Spreads centrifugally, with partial central clearing and a slightly elevated, erythematous or hyperpigmented, and sharply demarcated border.
- ☑ Infection may spread to the perineum and perianal areas, into the gluteal cleft, or onto the buttocks. In males, the scrotum is typically spared.

Differential Diagnosis

- ☑ *Candidal intertrigo:* Inflamed patches with satellite papules and pustules. Scrotal involvement is common.
- ☑ *Seborrheic dermatitis:* Moist, erythematous, or hyperpigmented patches in the genitocrural area.
- ☑ *Inverse psoriasis:* Smooth and shiny plaques with absent or minimal scale in intertriginous areas.
- ☑ *Erythrasma:* Erythrasma is a superficial bacterial infection of the skin caused by *Corynebacterium minutissimum*. Intertriginous involvement may present as erythematous to brown patches or thin plaques.

Tinea Unguium (Onychomycosis)

It affects nail plate.

Characteristic Lesion

- ☑ Superficial forms manifests as irregular single or numerous white patches on the surface of the nail not associated with deep paronychial infection.
- ☑ Invasive form manifests as subungual infection initiated at the lateral distal margins of the nail often preceded by mild paronychia. Nail becomes yellowish, thickened, brittle, and loosened from nail bed. In advanced cases, nail turns dark brown and cracks or breaks off.

Tinea Unguium (Onychomycosis)

Treatment

- ☑ Topical antifungal like ciclopirox without matrix involvement.
- ☑ Oral itraconazole—5 mg/kg/day for 1 week of each month for 3–4 months.
- ☑ Oral terbinafine at 62.5 mg/day (<20 kg), 125 mg/day (20–40 kg), and 250 mg (>40 kg) for 6 weeks for fingernails and 12 weeks for toenails.
- ☑ Oral griseofulvin (microsized) at 10–20 mg/kg/day for 6 months for fingernails and 6–12 months for toenails.

Tinea Barbae

- ☑ Tinea barbae is a dermatophyte infection involving beard hair in adolescent and adult males.
- ☑ Oral antifungal therapy is necessary.

Further Reading

- ☑ Goldstein AO, Goldstein BG. (2019). Dermatophyte (Tinea) Infections. [online] Available from https://www.studocu.com/en-gb/document/nottingham-trent-university/biomedical-science/dermatophyte-tinea-infections-up-to-date/24717319. [Last accessed November, 2022].
- ☑ Hsu S, Le EH, Khoshevis MR. Differential diagnosis of annular lesions. Am Fam Physician. 2001;64(2): 289-96.

Indian Academy of Pediatrics (IAP)

STANDARD TREATMENT GUIDELINES 2022

Tropical Pulmonary Eosinophilia

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